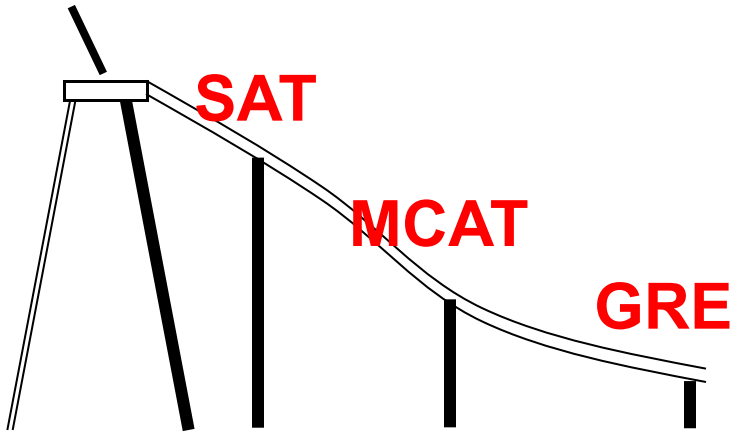
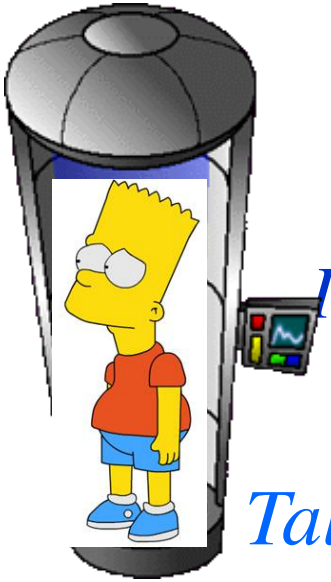
Test Slide



When Is Teleportation Quantum?
(and other musings on the relationship between physics and computation)



Dave Bacon

Department of Physics

Institute for Quantum Information

Caltech



Talk given at the Perimeter Institute

Waterloo, Ontario, Jan 21 2003







Canadian Bacon



Combining hockey with kung-fu

In Collaboration With…

Ben Toner

graduate student (advisor: John Preskill)

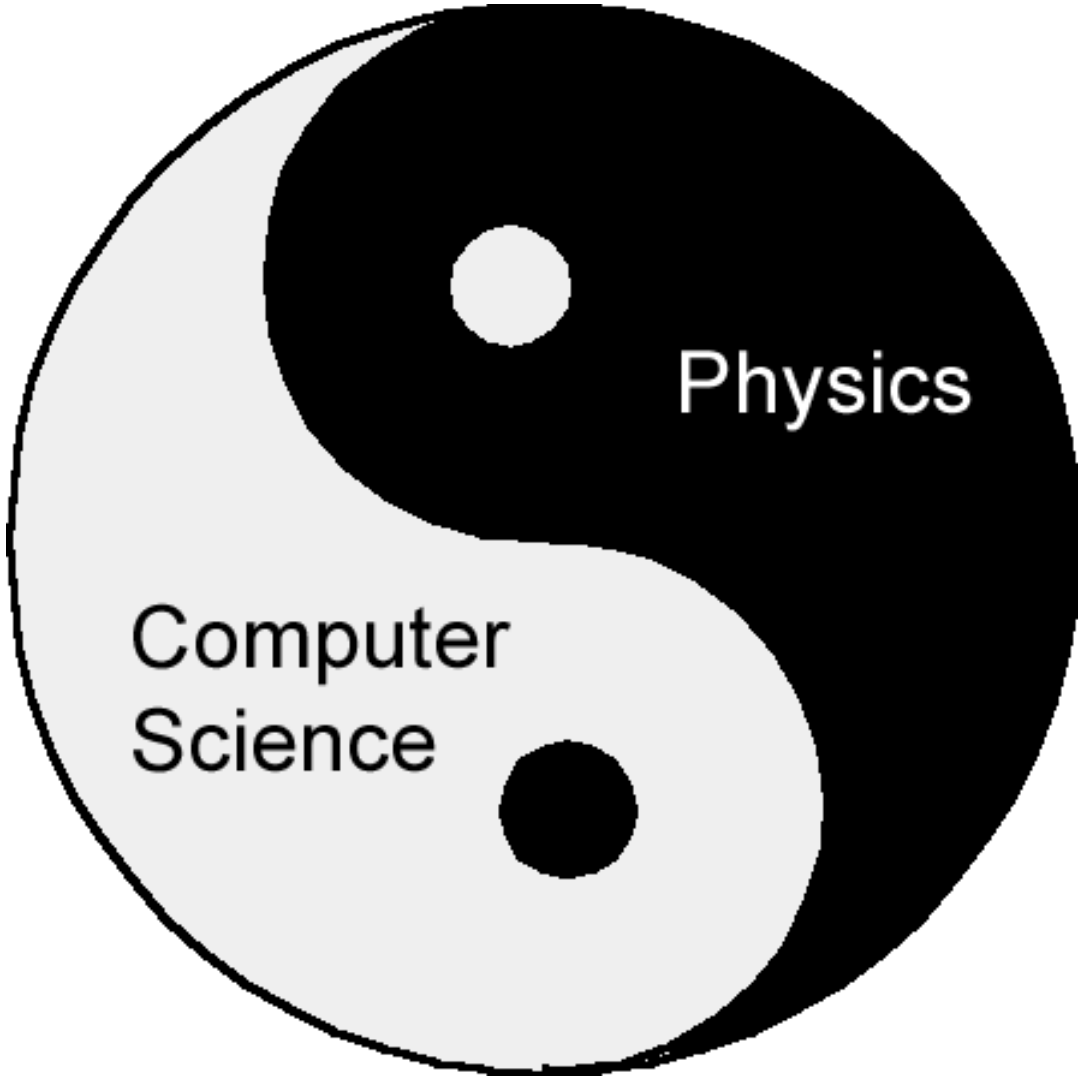
Department of Physics

Institute for Quantum Information

Caltech



Computers Obey Physics



Physics As Computation (?)

Computers Obey Physics

Quantum Computers



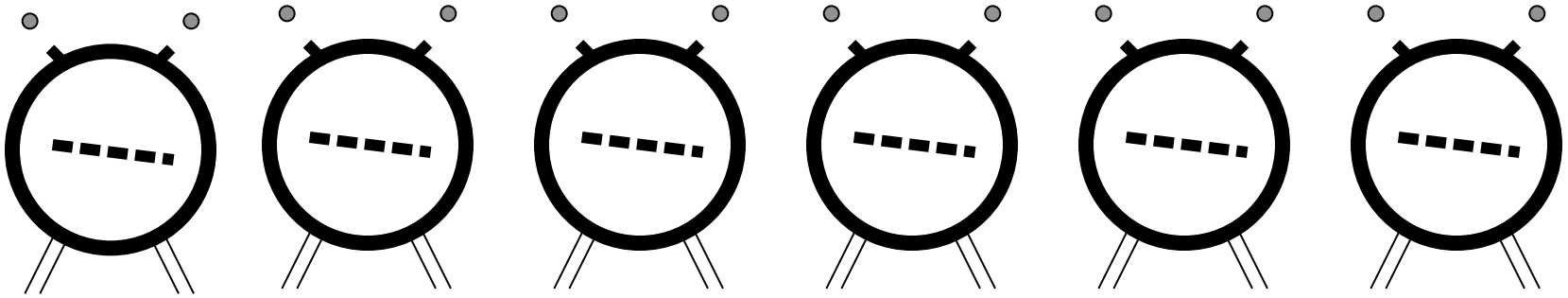
$$\frac{\sqrt{3}}{2}|\text{Intel}\rangle + \frac{1}{2}|\text{IBM}\rangle$$

Physical Theories imply computational complexity classes?

(locality?, locality!, locality?)

Economics of Information Technology

Physics As Computation



The Universe as a Quantum Computer?

Quantum theory as a theory of our information about reality.

Computer Science can motivate questions about Physics

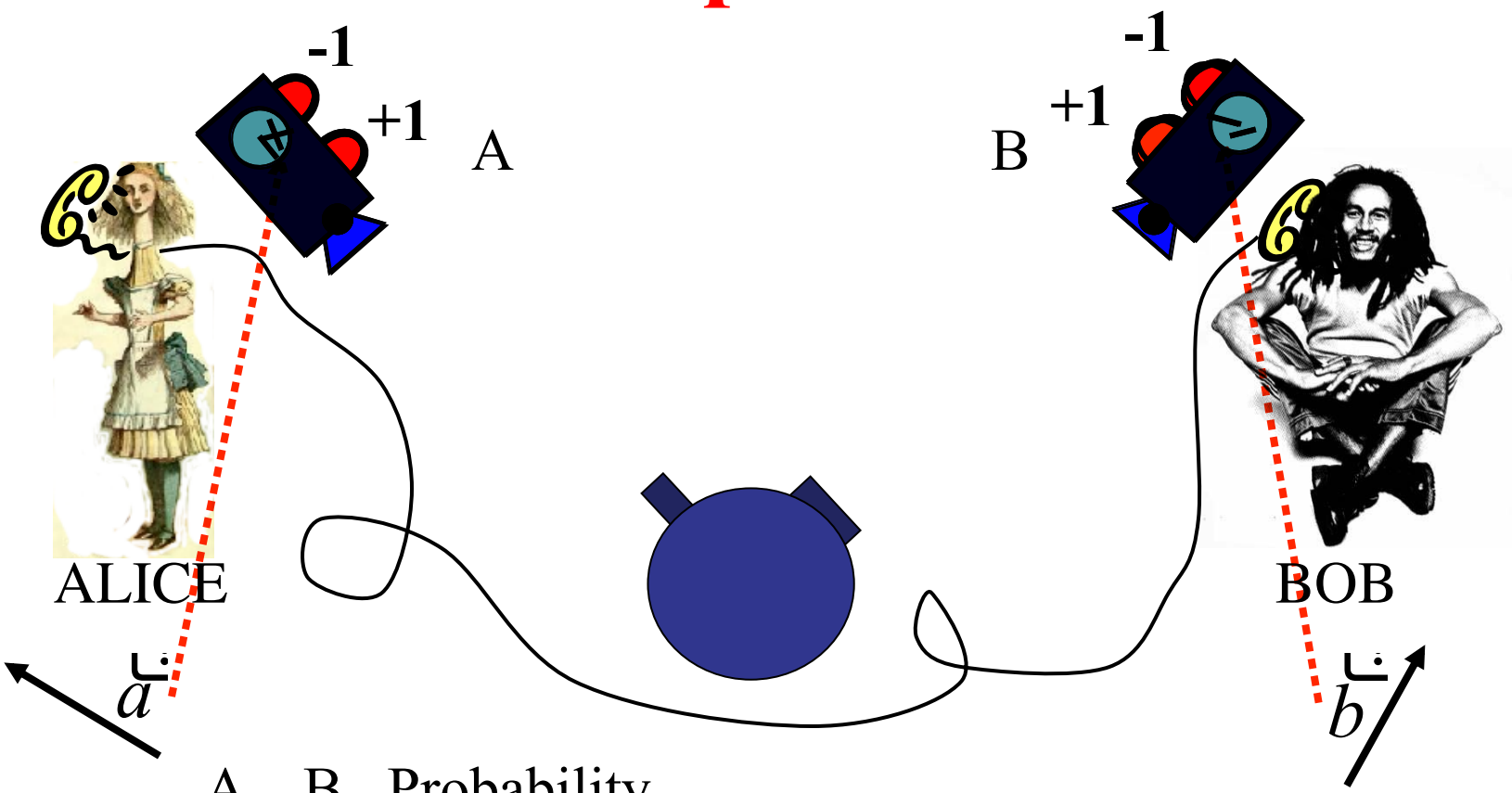
What Can We Learn About Physics From Theoretical Computer Science?

Silly Futurist Prediction?

Theoretical Computer Science, especially ideas originating in Quantum Information Science will greatly impact Theoretical Physics

In the Beginning There Was…

Bell Experiment



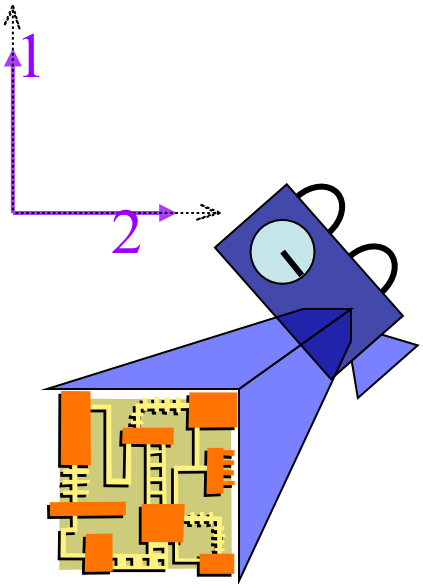
A	B	Probability
- 1	- 1	$\frac{1}{4}(1-\vec{a}\cdot\vec{b})$
+1	+1	$\frac{1}{4}(1-\vec{a}\cdot\vec{b})$
- 1	+1	$\frac{1}{4}(1+\vec{a}\cdot\vec{b})$
+1	- 1	$\frac{1}{4}(1+\vec{a}\cdot\vec{b})$

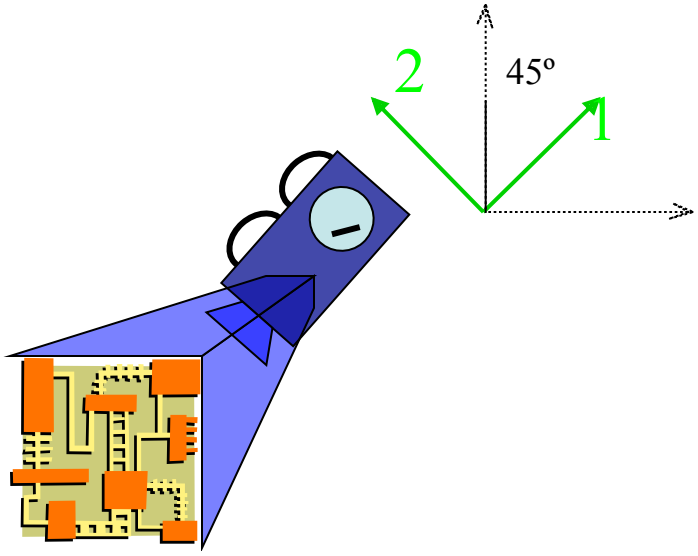
$$\langle A \rangle = \langle B \rangle = 0$$

$$\langle AB \rangle = -\vec{a}\cdot\vec{b}$$

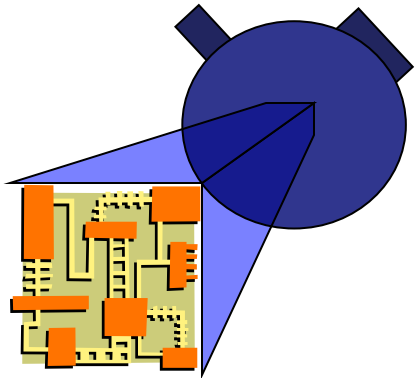
$$\langle AB \rangle = -\vec{a} \cdot \vec{b}$$

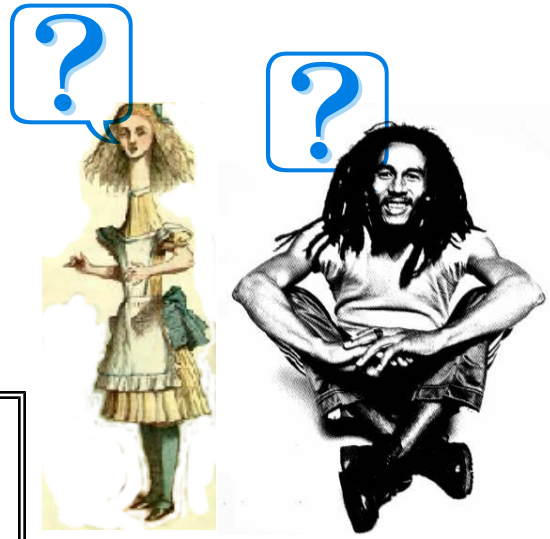
	1	2
1	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$
2	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$





?

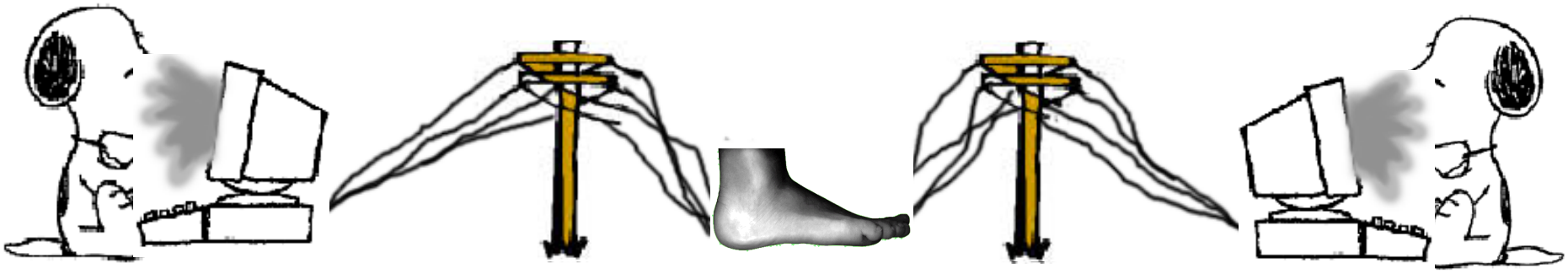




ALICE BOB

"Can a quantum system be probabilistically simulated by a classical (probabilistic, I'd assume) universal computer?" - R.P. Feynman.

Communication Breakdown



1. Initial unlimited classical communication between computers.
2. Communication lines cuts.
3. Input into each computer.
4. Output by each computer.

Repeat

Build up table of input-output correlations

What sort of correlations can be produced?

Bell Inequalities



Bell inequalities yield restrictions on correlations.

$$|\langle A_1 B_1\rangle + \langle A_1 B_2\rangle + \langle A_2 B_1\rangle - \langle A_2 B_2\rangle| \leq 2$$

A_i, B_i are ± 1 valued observables

Bell inequalities answer an information theoretic question:

What correlations can be produced using shared randomness and no communication between parties?

Make replacement

1. Initial unlimited quantum communication between computers.

   $\rightarrow$ parties can share entangled states.

$$|\langle \mathbf{A_1 B_1}\rangle + \langle \mathbf{A_1 B_2}\rangle + \langle \mathbf{A_2 B_1}\rangle - \langle \mathbf{A_2 B_2}\rangle| = 2\sqrt{2}$$

Capitalism for Correlations

Shared randomness
Communication
Quantum correlations
Super-quantum correlations



Shared randomness
No communication
Classical correlations

Bell inequalities

How much communication is necesary to produce quantum correlations?

What are the COSTS of producing quantum correlations?



What Can We Do?



1. Initial unlimited classical communication between computers.
2. Communication lines cuts.
3. Input into each computer.

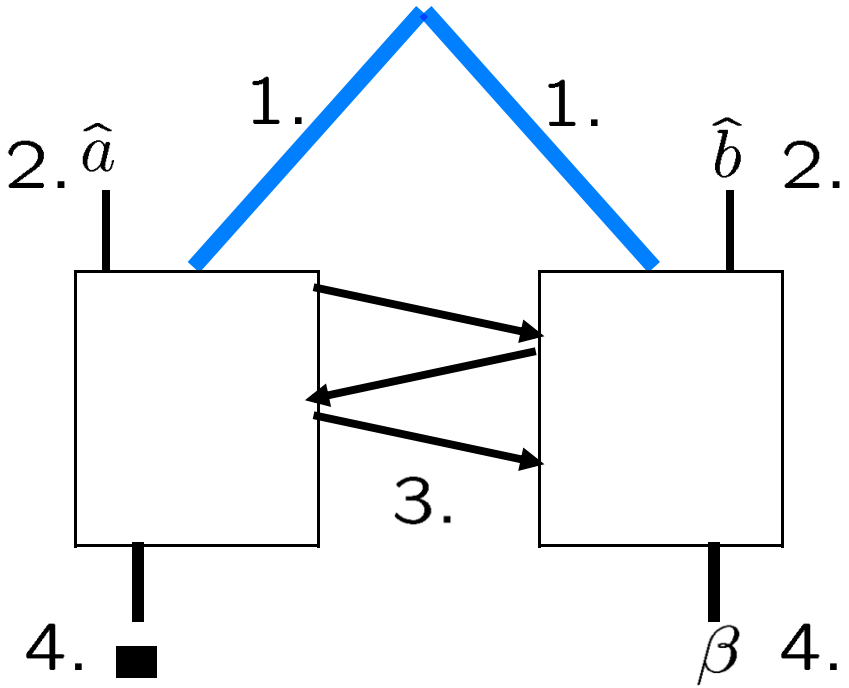
3.5. Metered rounds of communication

4. Output by each computer.

Repeat

What correlations can be produced using X amount of communication?

Simulating Bell Correlations



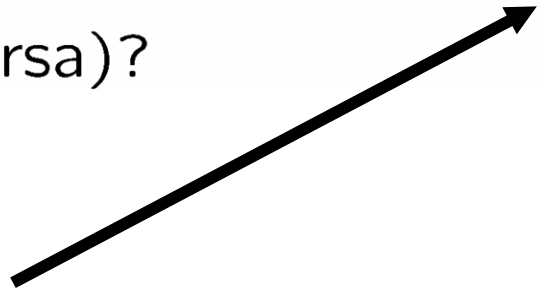
1. distribute shared random bits
2. A and B choose measurement directions, $\hat{a}$, $\hat{b}$
3. A and B communicate b bits
4. A and B output $\alpha, \beta \in \{\pm 1\}$

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta\hat{a}\cdot\hat{b}\right)$$

Why Is It Hard?

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right)$$

How to produce distribution without sending $\hat{a}$ from A to B (or vice versa)?



requires infinite number of bits

Can be done with 8 bits + shared randomness

Brassard, Cleve, Tapp (1999)

8 bits?

Towards Simulating Bell

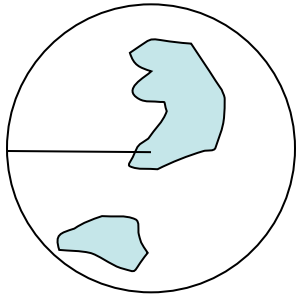
$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right)$$

Notice:

1. If $\hat{a} = \hat{b}$, then $Pr(\alpha = \beta) = 0$

   If $\hat{a} = -\hat{b}$, then $Pr(\alpha = \beta) = 1$

If B learns from A that A's vector lies in some region



and B knows A's output in this region

$\Downarrow$

forces B's output

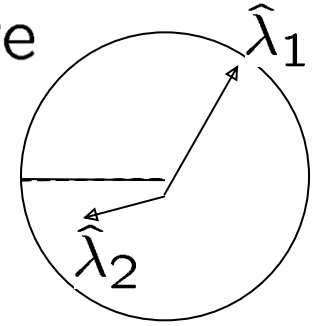
2. $\Pr\left(\alpha, \beta | \hat{a}, \hat{b}\right)$ is symmetric under joint rotations.

$$\Pr\left(\alpha, \beta | \hat{a}, \hat{b}\right) = \Pr\left(\alpha, \beta | R(\hat{a}), R(\hat{b})\right)$$

A 1 Bit Protocol

Two shared random variables:

$\hat{\lambda}_1, \hat{\lambda}_2$ uniformly distributed on sphere



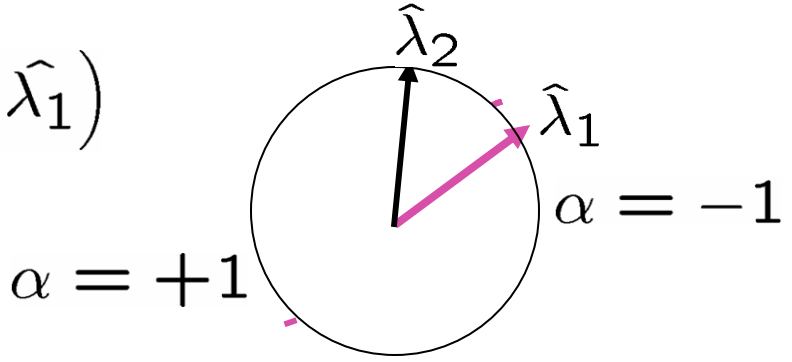
1. Alice outputs $-\text{sgn}\left(\hat{a} \cdot \hat{\lambda_2}\right)$

2. Alice sends a bit to Bob: $c = \text{sgn}\left(\hat{a} \cdot \hat{\lambda}_1\right) \text{sgn}\left(\hat{a} \cdot \lambda_2\right)$

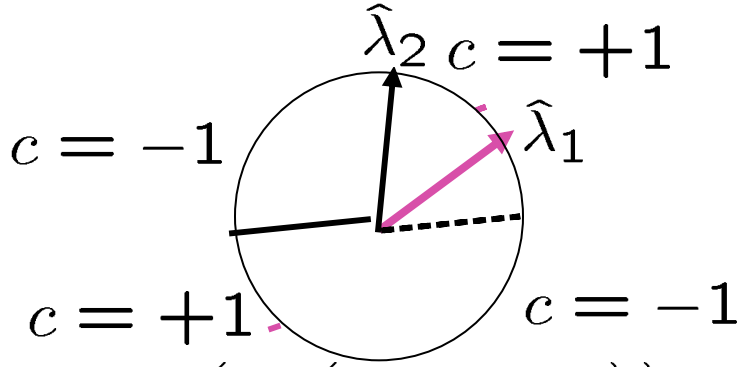
3. Bob outputs $\text{sgn}\left(\hat{b} \cdot \left(\hat{\lambda_1} + c\hat{\lambda_2}\right)\right)$

The Intuition

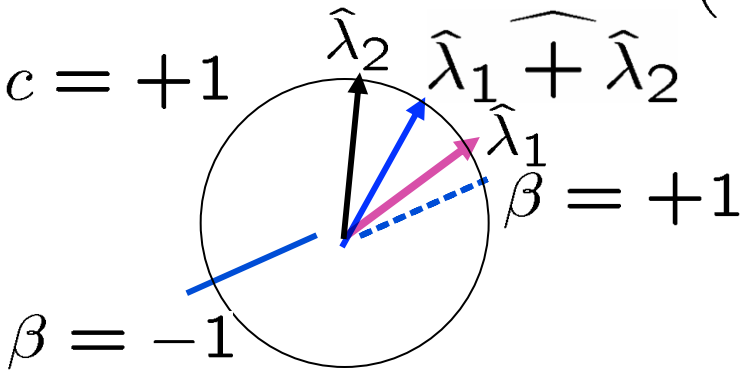
1. Alice outputs $-\text{sgn}\left(\hat{a}\cdot\hat{\lambda_1}\right)$

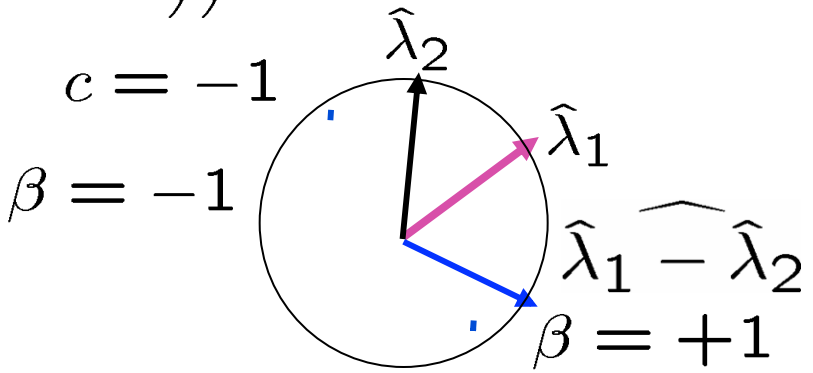


2. Alice sends a bit to Bob: $c=\text{sgn}\left(\hat{a}\cdot\hat{\lambda}_1\right)\text{sgn}\left(\hat{a}\cdot\lambda_2\right)$



3. Bob outputs $\text{sgn}\left(\hat{b}\cdot\left(\hat{\lambda_1}+c\hat{\lambda_2}\right)\right)$





It Works!

Each parties output changes sign under

$$\hat{\lambda}_1 \to -\hat{\lambda}_1,\ \hat{\lambda}_2 \to -\hat{\lambda}_2 \quad \Rightarrow \langle\alpha\rangle = \langle\beta\rangle = 0$$

$$\langle\alpha\beta\rangle = E\left[-\mathrm{sgn}(\hat{a}\cdot\hat{\lambda}_1)\sum_{d=\pm 1}\frac{1+cd}{2}\,\mathrm{sgn}\left[\hat{b}\cdot(\hat{\lambda}_1+d\hat{\lambda}_2)\right]\right]$$

$$E[x] = \frac{1}{(4\pi)^2}\int d\hat{\lambda}_1\int d\hat{\lambda}_2[x] \qquad c = \mathrm{sgn}\left(\hat{a}\cdot\hat{\lambda}_1\right)\mathrm{sgn}\left(\hat{a}\cdot\lambda_2\right)$$

$$\langle\alpha\beta\rangle = E\left[-2\,\mathrm{sgn}(\hat{a}\cdot\hat{\lambda}_1)\,\mathrm{sgn}\left[\hat{b}\cdot(\hat{\lambda}_1+\hat{\lambda}_2)\right]\right]$$

$$\frac{1}{4\pi}\int d\hat{\lambda}_2 \mathrm{sgn}\left(r+\hat{b}\cdot\hat{\lambda}_2\right) = r$$

$$-\frac{1}{4\pi}\int d\hat{\lambda}_1 2\mathrm{sgn}\left(\hat{a}\cdot\hat{\lambda}_1\right)\hat{b}\cdot\hat{\lambda}_1 = -\hat{a}\cdot\hat{b}$$

$$\langle\alpha\beta\rangle = -\hat{a}\cdot\hat{b}$$

1 Bell Pair's Worth of Correlations

correlations from projective measurements on Bell pair

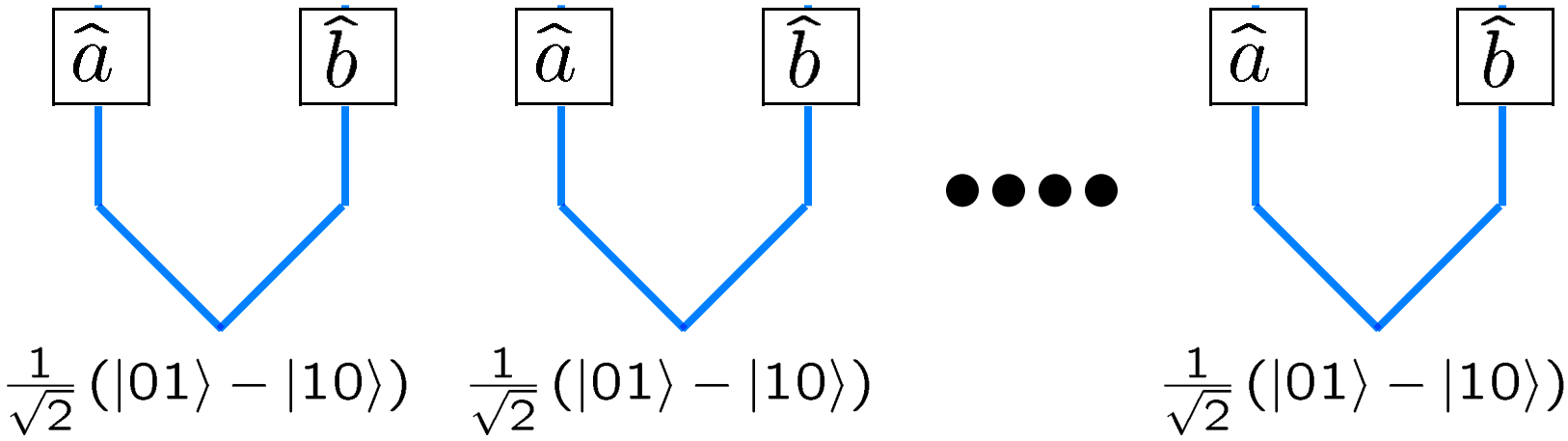
$=$

shared randomness

$+$

1 bit of communication

Average Communication



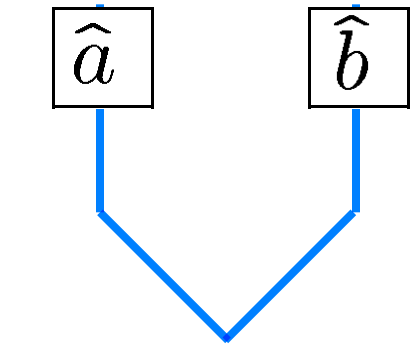
Simulate the correlations for n Bell pair experiments.

Can compress amount of communication to $\approx 0.85n$ bits.

1 Bell Pair's Worth of Correlations Implications

1. Produces "Detector Inefficiency" local hidden variable model.

Connection to Detector Inefficiency



1. Alice outputs
2. Alice sends bit to Bob
3. Bob outputs

Distribute extra random bit, 50% 0, 50% 1

A and B run protocol as if random bit was communicated bit

If A finds that she was going to send a bit $\neq$ random bit,

Alice says: "my detector didn't fire"

If A finds that she was going to send a bit $=$ random bit,

Great. Protocol has produced correlations

Symmetrize + remove detector correlations



local hidden variable model
with detector inefficiency

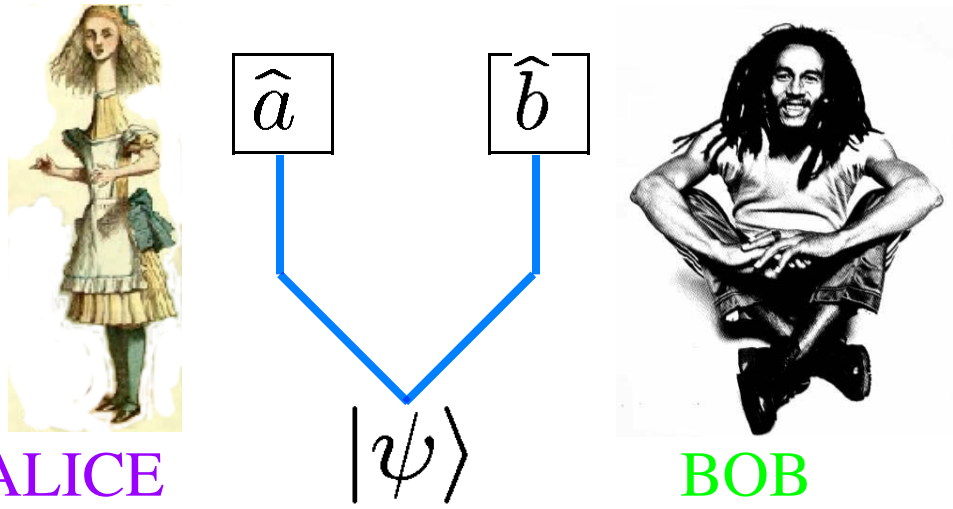
Computer Science Giving Back To Physics

1 Bell Pair's Worth of Correlations Implications

1. Produces "Detector Inefficiency" local hidden variable model.

2. Simulating all two outcome correlation observables with bounded communication

Correlation Observable



$d_A d_B$ dimensional state pure state

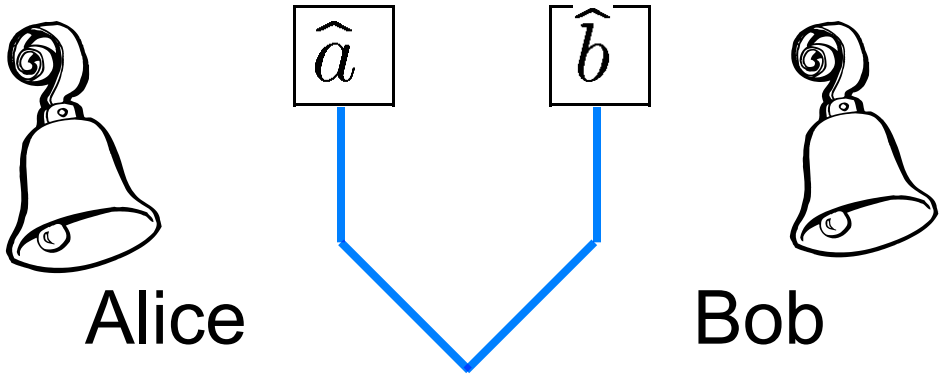
$\mathbf{A}, \mathbf{B}$ are ± 1 valued observables $\quad \mathbf{A}^2 = \mathbf{B}^2 = \mathbf{I}$

$$\langle \mathbf{A} \otimes \mathbf{B} \rangle = \hat{a} \cdot \hat{b}$$

$\hat{a}$, $\hat{b}$ are now $2d^2$ dimensional unit vectors, $d = \max(d_A, d_B)$

Generalized Bell Correlations

A's output: $\alpha \in \pm 1$ $\beta \in \pm 1$:Bob's output



$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right)$$

$\hat{a}$, $\hat{b}$ are now d dimensional unit vectors.

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right) \Rightarrow \langle \mathbf{A} \otimes \mathbf{B} \rangle = \hat{a} \cdot \hat{b}$$

Chain Letter

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right) \quad (\hat{a}, \hat{b} \text{ d dimensional})$$

+

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right) \quad (\hat{a}, \hat{b} \text{ 3 dimensional})$$

=

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right) \quad (\hat{a}, \hat{b} \ d+2 \text{ dimensional})$$

$\langle \mathbf{A} \otimes \mathbf{B} \rangle = \hat{a} \cdot \hat{b}$ using d^2 bits of communication

(for any mixed state $\boldsymbol{\rho}$)

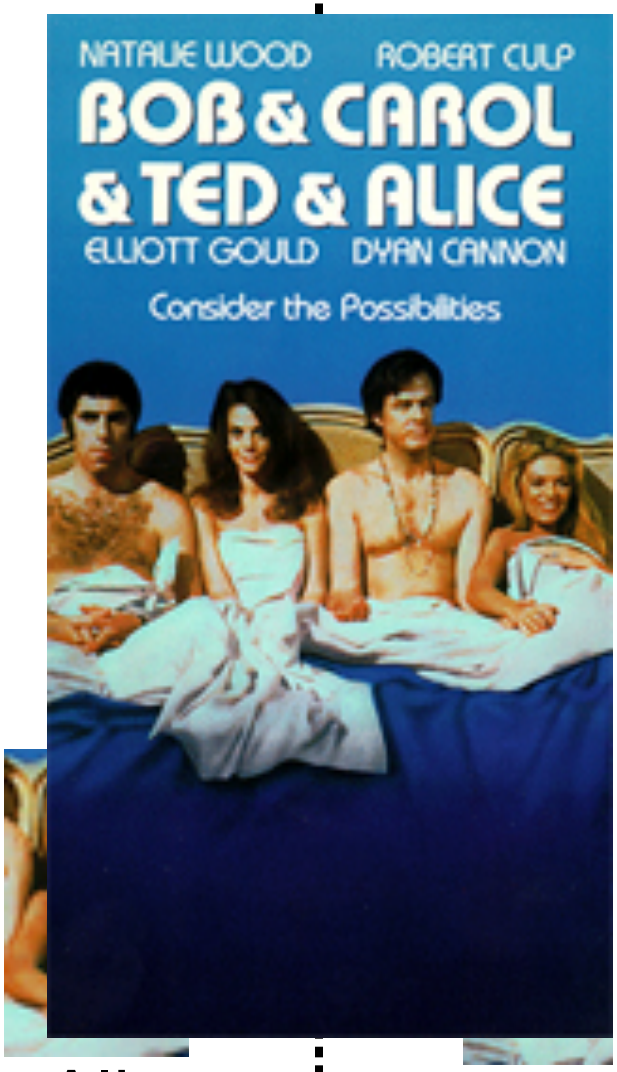
1 Bell Pair's Worth of Correlations Implications

1. Produces "Detector Inefficiency" local hidden variable model.

2. Simulating all two outcome correlation observables with bounded communication

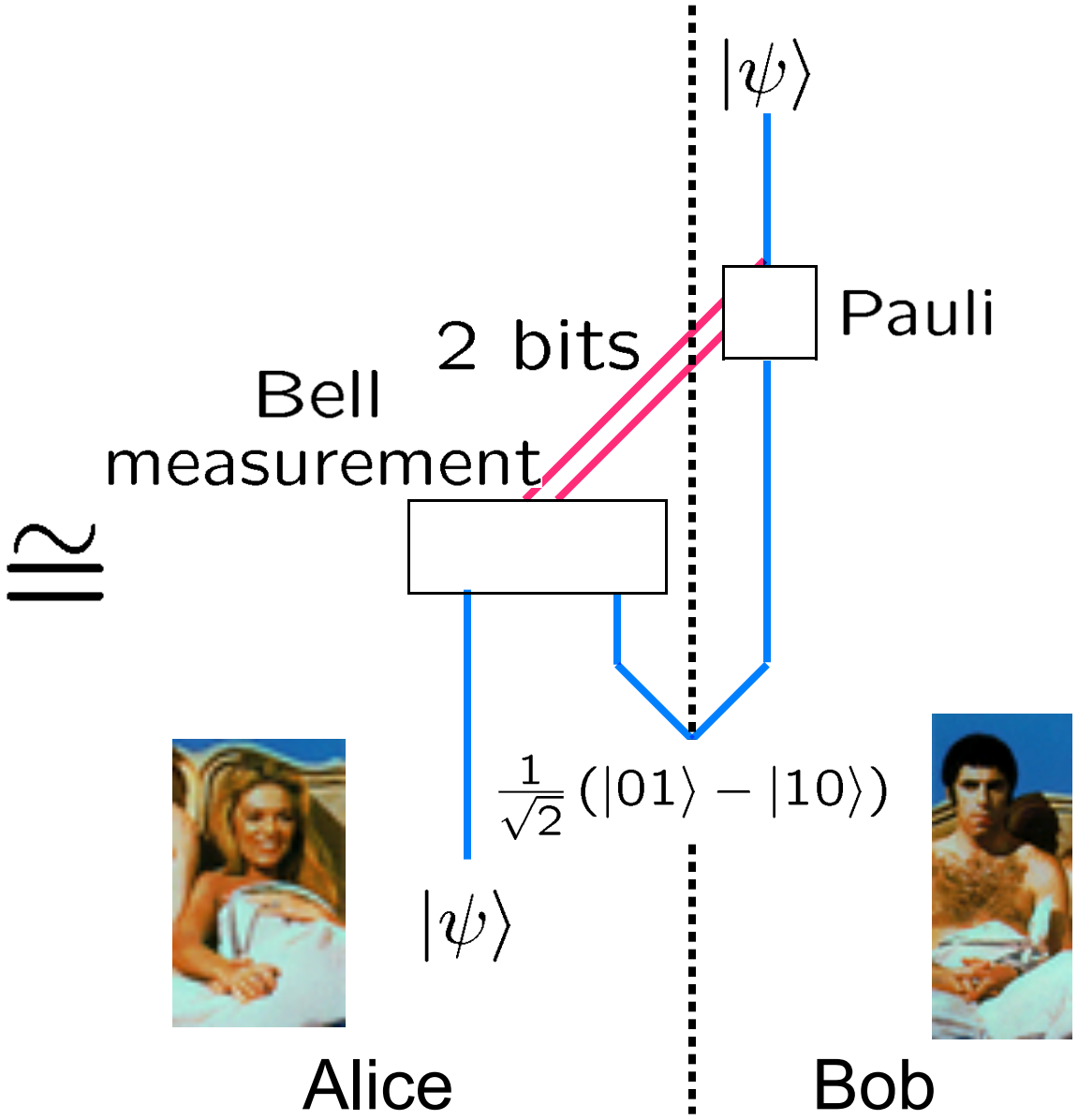
3. Helps delineate "when quantum teleportation is really quantum"

Teleportation



Alice Bob

$\cong$



Alice Bob

An Experiment

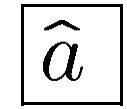
"qubit teleporation"



Alice



Bob



$\hat{b}$ measurement $\mathbf{P}_{\pm} = \frac{1}{2}\left(\mathbf{I} \pm \hat{b} \cdot \hat{\boldsymbol{\sigma}}\right)$

Prob(+1: along $\hat{b}$)$=\frac{1}{2}\left(1 + \hat{a} \cdot \hat{b}\right)$

Prob(-1: along -$\hat{b}$)$=\frac{1}{2}\left(1 - \hat{a} \cdot \hat{b}\right)$

2 bits

$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

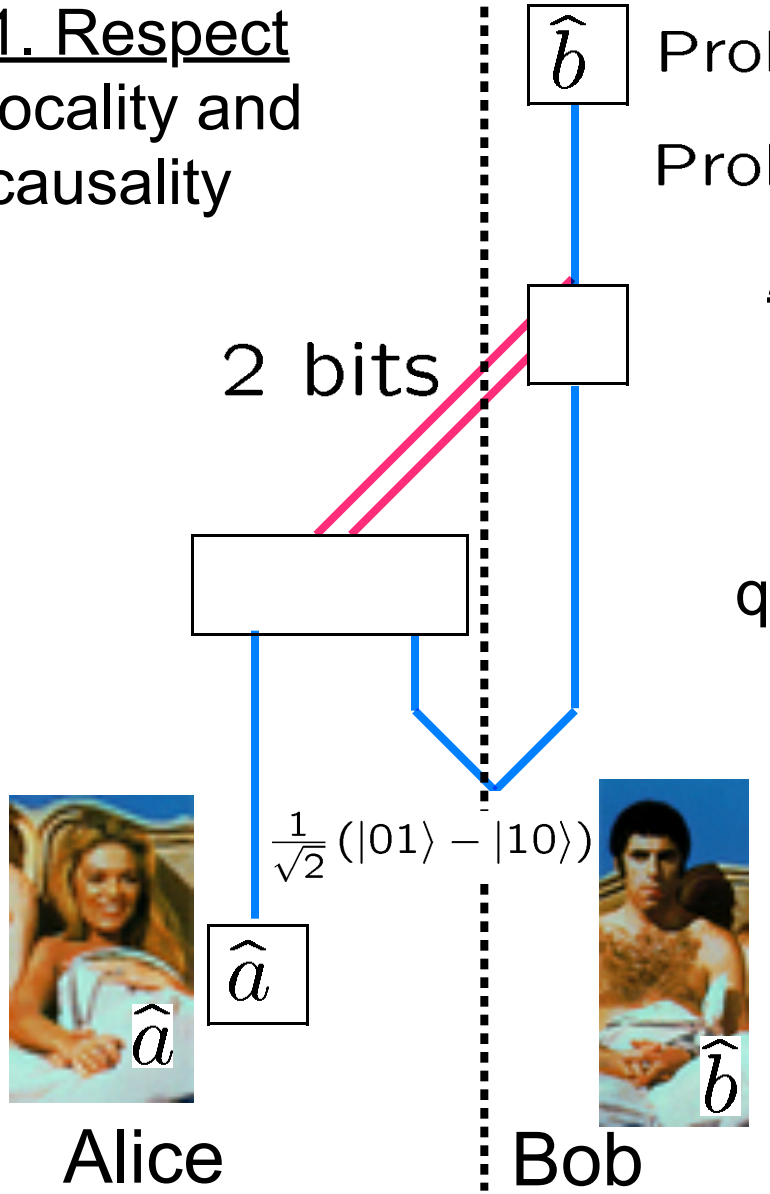
preparation $|\psi\rangle$

$\rho = \frac{1}{2}(\mathbf{I} + \hat{a} \cdot \hat{\boldsymbol{\sigma}})$

In what sense is this experiment "quantum"?

Simulating Classically

1. Respect locality and causality



$$\text{Prob}(+1\text{: along } \hat{b}) = \frac{1}{2}\left(1 + \hat{a}\cdot\hat{b}\right)$$

$$\text{Prob}(-1\text{: along } -\hat{b}) = \frac{1}{2}\left(1 - \hat{a}\cdot\hat{b}\right)$$

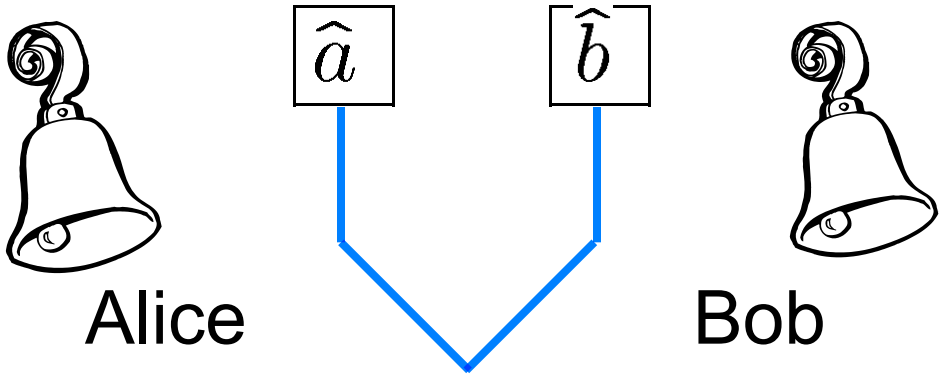
2. replace

quantum wire by classical wire carrying any amount of classical info

classical wire b bits by classical wire b bits

Simulating Bell Correlations

A's output: $\alpha \in \pm 1$ $\beta \in \pm 1$:Bob's output



$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right)$$

simulating Bell Correlations with b bits communication and shared randomness $\Longrightarrow$ simulating qubit teleportation with b+1 bits communication and shared randomness

"Classical teleportation of a qubit" Cerf, Massar, Gisin (2000)

Bell Pair to Qubit Teleportation

Alice and Bob can produce

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right)$$

Alice sends her output bit (α) to Bob

If Alice's output bit equals Bob's output bit,
Bob outputs -1
Otherwise
Bob outputs +1

qubit teleportation

$$\Pr(\alpha = \beta | \hat{a}, \hat{b}) = \frac{1}{2}\left(1 - \hat{a} \cdot \hat{b}\right) \quad \Pr(\text{B outputs } -1) = \frac{1}{2}\left(1 - \hat{a} \cdot \hat{b}\right)$$

$$\Pr(\alpha \neq \beta | \hat{a}, \hat{b}) = \frac{1}{2}\left(1 + \hat{a} \cdot \hat{b}\right) \quad \Pr(\text{B outputs } +1) = \frac{1}{2}\left(1 + \hat{a} \cdot \hat{b}\right)$$

Qubit Teleportation

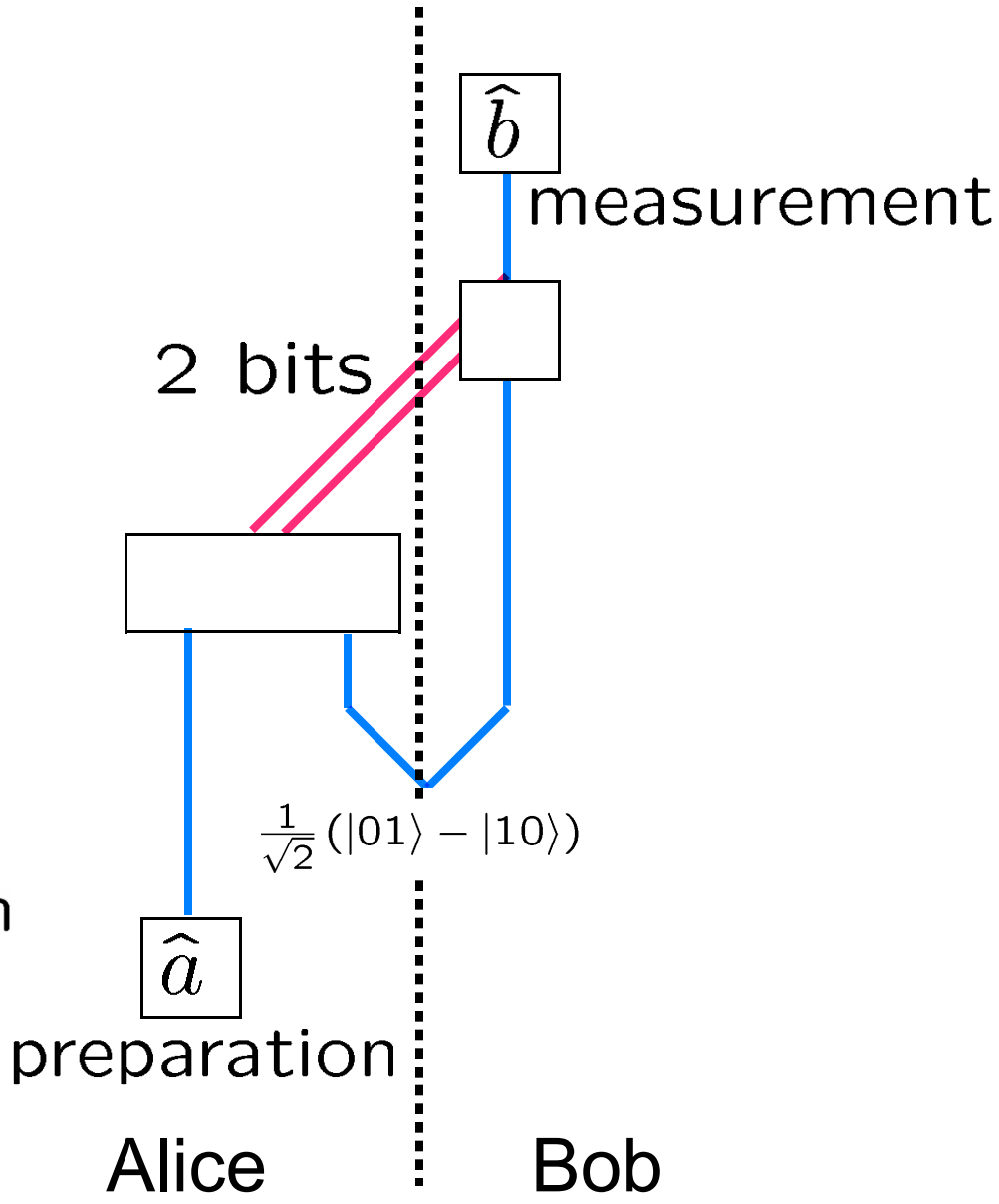
qubit teleportation

=

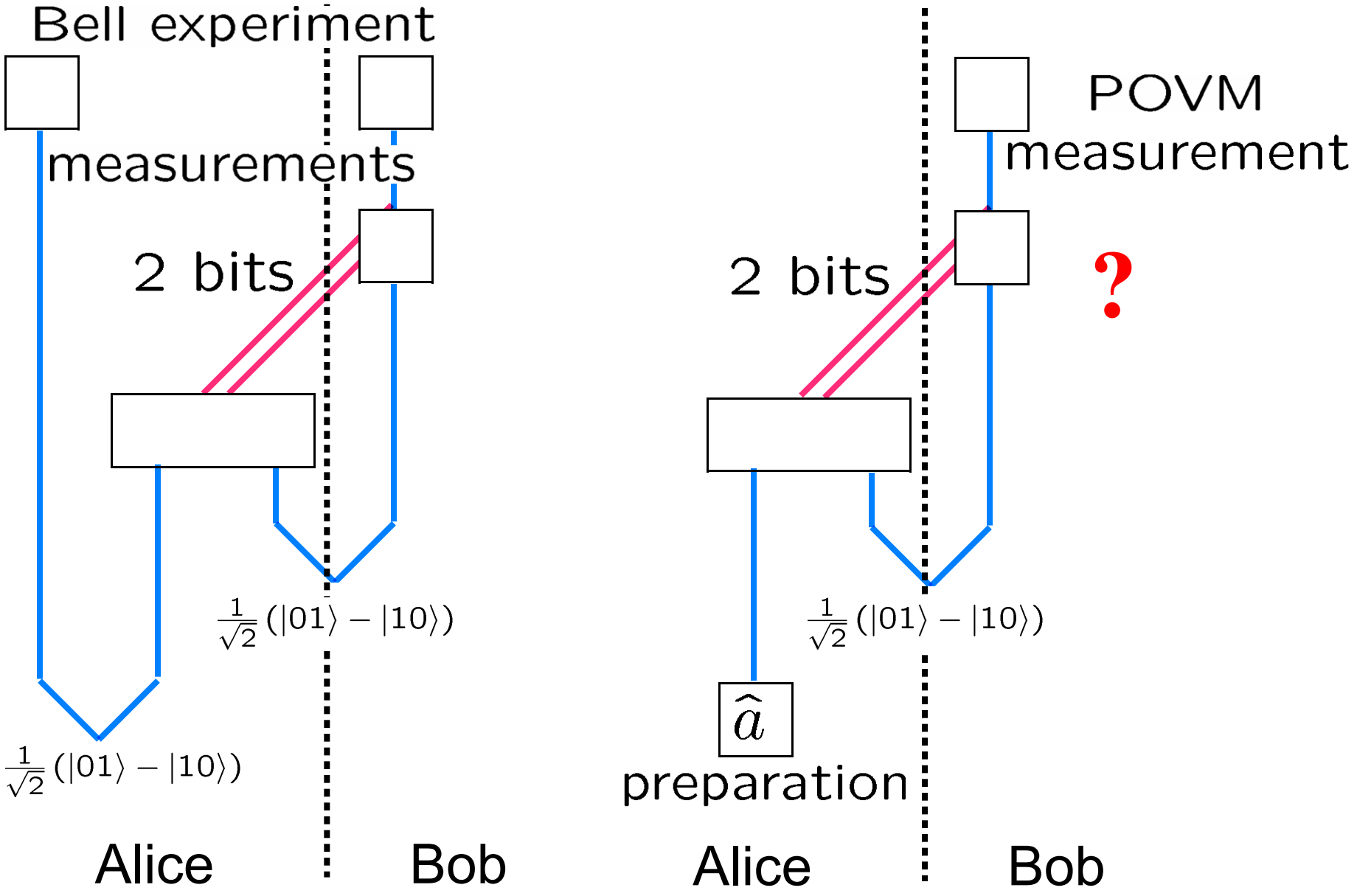
shared randomness

+

2 bits of communication



Quantum Teleportation



1 Bell Pair's Worth of Correlations Implications

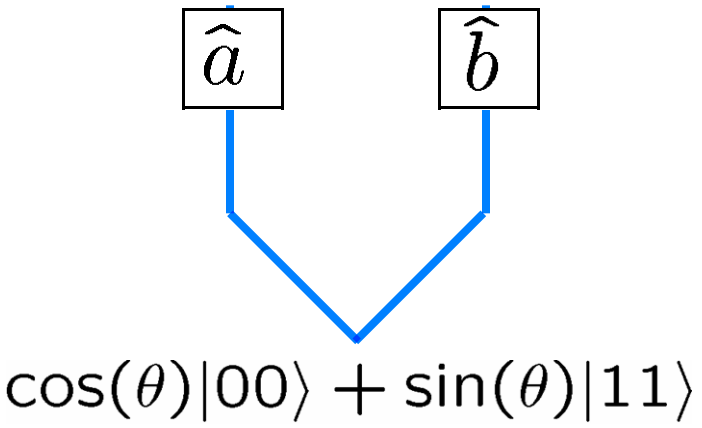
1. Produces "Detector Inefficiency" local hidden variable model.

2. Simulating all two outcome correlation observables with bounded communication

3. Helps delineate "when quantum teleportation is really quantum"

4. Simulating projective measurements on non maximally entangled two qubit states with two bits of communication

Correlations on Non-maximally Entangled 2 Qubit States



Alice simulates measurement along $\hat{a}$, output α

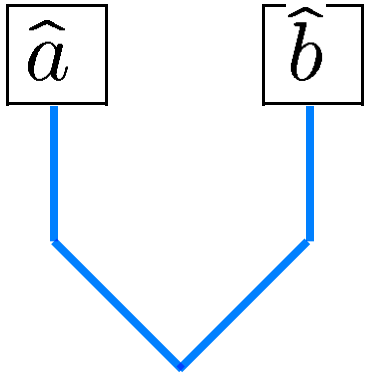
Alice now can calculate what state Bob should have:

Call state $\hat{a}'$

Parties run qubit teleportation scheme to make Bob measure $\hat{a}'$ along $\hat{b}$, output β

2 bits

A Cryptographic Interpretation of Quantum Theory?



entanglement

shared random key

+

finite bits of nonlocal communication

Nonlocal theories need to explain why only correlations between entangled parties exist?

References

Bacon and Toner, “Bell Inequalities with Communication” PRL **90** 157904 (2003)

Toner and Bacon, “The Communication Cost of Quantum Correlations” PRL **91** 187904 (2003)

Bacon and Toner, “The Communication Cost of Simulating Joint Observable Correlations” *in preparation* (2004)

Open Problems

Generic measurements on generic states?

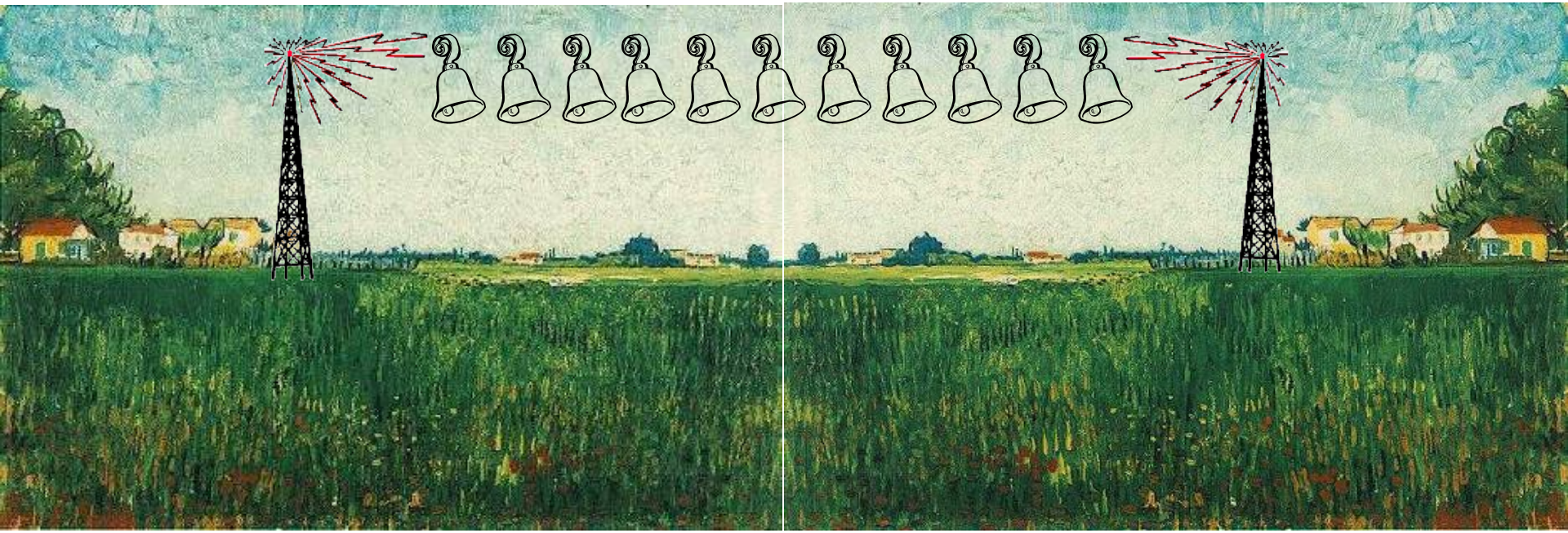
What is THE GAME?

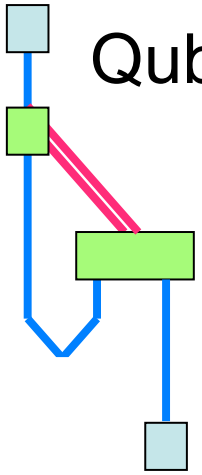
Quantify the "other" quantum phenonmenon: Kochen-Specker

Find a Bell inequality with communication which is violated by quantum mechanics

Fin

Correlations on Bell Pair = shared randomness + 1 bit





Qubit teleportation can be simulated classically!



